

Artificial social influence via human-embodied AI agent interaction in immersive virtual reality (VR): Effects of similarity-matching during health conversations

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ABSTRACT

Interactions with artificial intelligence (AI) based agents can positively influence human behavior and judgment. However, studies to date focus on text-based conversational agents (CA) with limited embodiment, restricting our understanding of how social influence principles, such as physical similarity, apply to AI agents (i.e., artificial social influence). We address this gap by leveraging latest advances in AI (large language models) and combining them with immersive virtual reality (VR). Specifically, we built VR-ECAs, or embodied conversational agents that can engage in turn-taking conversations with humans about health-related topics in a virtual environment. Then we manipulated interpersonal similarity via gender matching and examined its effects on biobehavioral (i.e., gaze), social (e.g., agent likeability), and behavioral outcomes (i.e., healthy snack selection). We observed an interaction effect between agent and participant gender on biobehavioral outcomes: discussing health with opposite-gender agents tended to enhance gaze duration, with the effect stronger for male participants compared to their female counterparts. A similar directional pattern was observed for healthy snack selection. In addition, female participants liked the VR-ECAs more than their male counterparts, regardless of the VR-ECAs' gender. Finally, participants experienced greater presence while conversing with embodied agents than chatting with text-only agents. Overall, our findings highlight embodiment as a crucial factor of AI's influence on human behavior, and our paradigm enables new experimental research at the intersection of social influence, human-AI communication, and immersive virtual reality (VR).

1. Introduction

The idea of intelligent machines that can exhibit human-like social behaviors has piqued the curiosity of many for decades (Vogele & Bente, 2010), but this idea long remained within the realm of dreams and speculation. Recent advances in artificial intelligence (AI), particularly in large language models (LLMs), as well as immersive virtual reality (VR) and avatar interfaces, now enable the creation of embodied conversational agents (VR-ECAs) that can engage in conversations that resemble natural human dialogue. Thus, by using VR to create environments in which humans can encounter and interact with LLM-based agents, who have a human-like appearance and exhibit believable behaviors, researchers can create a unique interface and paradigm to study human-AI communication. Specifically, by leveraging VR-ECAs,

researchers can better understand the social and biobehavioral processes underlying AI agents' influence on human judgment and behavior (i.e., artificial social influence).

This study introduces an innovative research paradigm that involves real-time conversations with VR-ECAs about health. Theoretically, we examine the effect of interpersonal similarity on participants' gaze towards the agent, evaluations of the agent (e.g., agent likeability), and behavior (i.e., snack selection). In this paper, we first introduce each aspect of our research paradigm. Then we build our research questions and hypotheses based on existing work. Next, we outline the methodology and present the results of our experimental manipulations. Finally, we discuss how the findings advance our understanding of artificial social influence and computer-mediated communication more broadly.

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¹ In the context of this study, the word agent is used broadly to refer to a non-human entity that responds to the environment "in pursuit of its own agenda" (Franklin & Graesser, 1996).

1.1. Embodied conversational AI: powerful agent of influence

AI refers to the field of study that aims to understand and build intelligent machines (Nilsson, 2009; Russell & Norvig, 2020), or systems that exhibit humans' cognitive capability to problem-solve, learn, and think (Holzinger et al., 2019). Previously, interactive systems such as text-based conversational agents (CAs) mimicked human-human conversations, but they had limited natural language understanding and generation abilities. However, recent LLM-based systems (e.g., OpenAI's ChatGPT) can process input from users and generate high-quality communicative content at a rapid speed. These systems also have capabilities beyond just producing text (Bubeck et al., 2023; Elyoseph et al., 2023; Wei et al., 2022). For instance, LLMs can generate clear and appropriate health promotion messages (Lim & Schmälzle, 2023) with greater argument strength and perceived effectiveness than humans including health experts (Karinshak et al., 2023; Lim & Schmälzle, 2024). AI can also influence individuals by catering responses to the receivers' personalities (Matz et al., 2024) and building relationships with the user (Burtell & Woodside, 2023).

Prior research on human-AI communication and AI-driven influence has primarily focused on interactions with text-based CAs, leaving a large gap in our understanding of artificial social influence. Text-based CAs offer purely language-based and written interaction and lack embodiment (e.g., a voice and a body). However, human cognition is heavily intertwined with the body (Barsalou, 2008; Shapiro et al., 2021; Wilson, 2002). We interact with the world - and with each other - via primary modalities like grasping, seeing, and hearing. Humans also rely heavily on paralanguage, kinesics, and nonverbal cues to communicate, receive feedback, form judgments, and make decisions (Burgoon et al., 2021; Hans & Hans, 2015). Furthermore, systems like gaze behavior and behavioral mimicry are powerful mechanisms that impact social (Chartrand & Lakin, 2013) and behavioral outcomes (Cialdini & Goldstein, 2004; Van Baaren et al., 2004) of social interactions.

Related works in human-agent interaction and affective computing also underscore the importance of embodiment and *non-verbal* aspects of the interaction (Cassell et al., 1999; De Rosi et al., 2003; Vinciarelli et al., 2011). For instance, Appel et al. (2012) found that participants responded more favorably while sharing stories with an embodied agent compared to sharing with a text-based interface. Krämer et al. (2007) and Gratch et al. (2006) showed that embodied agents' nonverbal cues (e.g., dynamic listening behaviors, self-touching gestures) significantly impacted participants' experience and post-interaction evaluations. Therefore, only focusing on text-based CAs overlooks how the embodiment features of CAs foster social connections with humans and exert social influences.

1.2. Virtual reality (VR): embodiment, immersive real-time interactions, and biobehavioral measurement

Immersive VR offers the capability to simulate real-time interactions between humans and VR-ECAs and rigorously examine the mechanism underlying artificial influence and social behavior in general. As mentioned above, the recent breakthroughs in VR and avatar-creation platforms now allow for the creation of VR-ECAs. These VR-ECAs can appear highly realistic (though other appearance features are also possible) and exhibit nonverbal behavior. When integrated with AI text-generation, text-to-speech, and speech-to-text capacities, such artificial human-like agents (Vogele & Bente, 2010) can mimic in-person human-human interactions in immersive virtual environments.

The concept of immersiveness refers to the capability of technology to absorb users' perceptual system, block out real-world sensory input, and stimulate it authentically with artificially generated simulacra (Bente et al., 2023; Biocca et al., 1995). VR's immersive nature thus offers the unique opportunity for humans to encounter artificial agents in the same virtual space and engage in verbal back-and-forth dialogue with them. With biobehavioral measures like eye-tracking, which are

increasingly integrated into head-mounted displays, we can now gather more precise information about the mechanisms of interpersonal interaction within virtual environments (Andrei et al., 2023; Syrjämäki et al., 2020). Indeed, scholars have already begun to examine how factors like the human-agent similarity influence the interaction in VR (Shih et al., 2023), and we expect a rapid expansion of VR-based research with AI agents.

1.3. Similarity: essential ingredient of artificial social influence

It is well-known that similarity among interacting individuals - both observable and perceived - has a powerful effect on human behavior. For instance, human relationships and connections are often formed based on shared attributes, including ethnic and cultural backgrounds, interests, and attitudes (i.e., homophily; McPherson et al., 2001). Rogers and Bhowmik (1970) additionally proposed that - when given a choice - people prefer to engage with and become more receptive to those who resemble them. Likewise, Cialdini and Goldstein (2004) suggested that during interactions with strangers, similarity acts as a heuristic for potential future friendship or acquaintance, enhancing the likelihood of compliance. Furthermore, social identity theory (SIT) posits that people are driven to categorize themselves and others into distinct groups based on factors such as gender and think and act according to the group they identify with (Ashforth & Mael, 1989; Hogg et al., 1995; Tajfel et al., 1985). Thus, messages from those categorized as similar (i.e., in-group members) tend to exhibit greater influence than messages from dissimilar individuals (i.e., out-group members; Wood, 2000). The influence of similarity has also been demonstrated in human-agent interaction literature (Jin & Eastin, 2023; Liao & He, 2020; Qiu & Benbasat, 2010).

1.4. Current study

Building on these findings, this study examined the effect of human-agent similarity on gaze during and evaluation of agent and healthy food choice after a dialogue about health in immersive VR (see Fig. 1). We specifically focused on gender matching, a salient characteristic in person perception and a key factor of embodiment. Gender matching, in this study, refers to the VR-ECA having the same gendered appearance as the human user. We chose gender because it is a basic indicator of similarity, especially during the first few encounters with the person. Those with the same gender share a similar bodily appearance and experience comparable societal expectations (controlling for drastic differences in ethnic identity).

To study the effect of gender matching, we created VR-ECAs by integrating OpenAI GPT4 and text-to-speech and speech-to-text AI models with avatars that displayed basic lip sync and gaze behavior in a VR platform. We also developed text-based CAs with no embodiment cues as the control condition. Since text-based CAs are the most common type of LLM-based system available, we used the control condition to examine whether immersive VR and embodiment influenced participants' perception of presence. Participants were randomly assigned to interact with either a gender-matched VR-ECA, a gender-unmatched VR-ECA, or a text-based CA.

2. Hypothesis and research questions

2.1. Effect of gender matching on biobehavioral, social, and behavioral outcomes

We first examined how gender matching influences gaze behavior, a key nonverbal interaction signal, during the human-agent interaction. It is often said that the eye is the window to the soul. Indeed, during social interactions, the gaze signals social attention (Holleman et al., 2020). Furthermore, there is a natural gaze pattern of adaptation and coordination (e.g., eye contact, then breaking eye contact) that occurs during

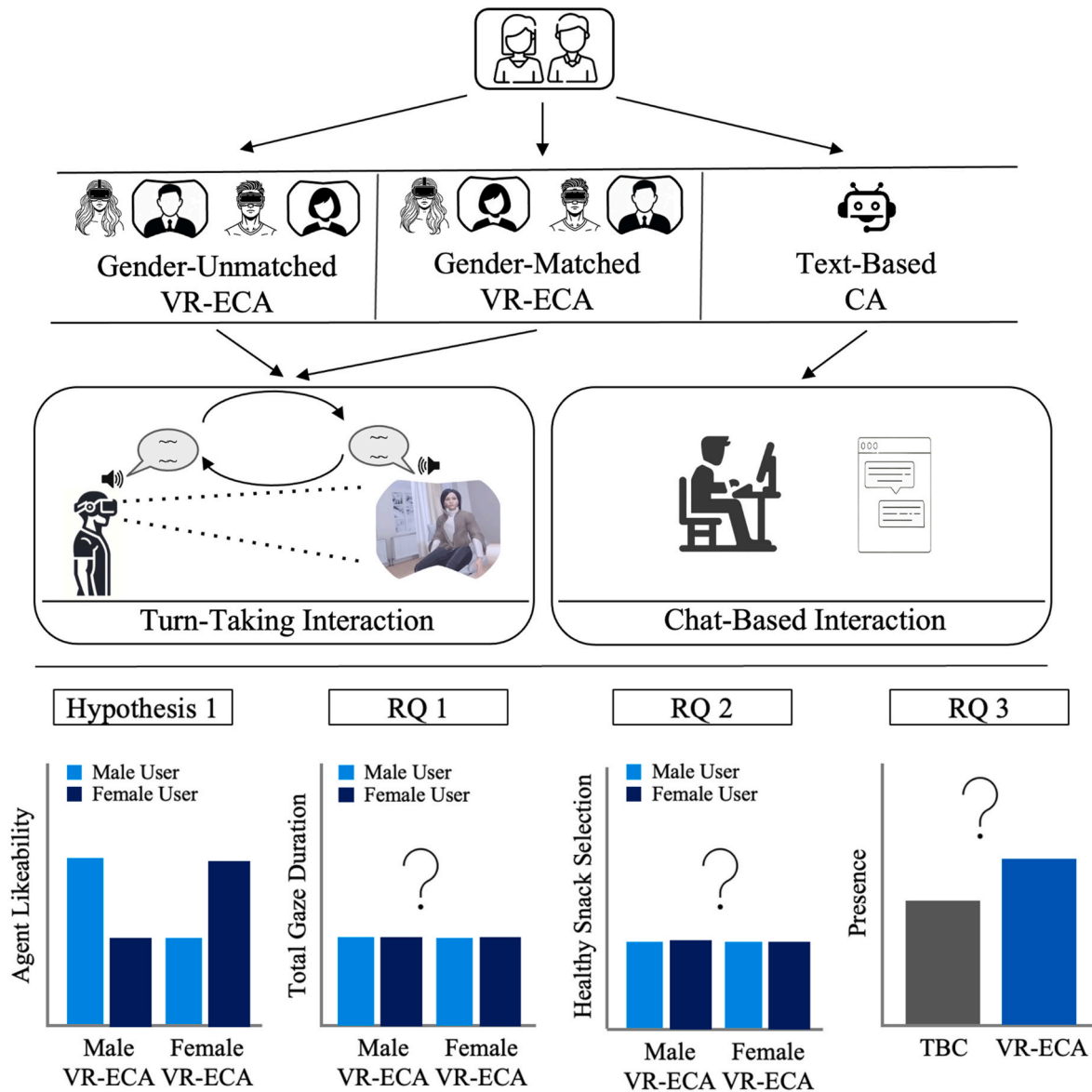


Fig. 1. Conceptual figure illustrating the study design and research question/hypotheses.

the conversation to send and receive signals such as attentiveness (Abele, 1986; Bente et al., 1998; Pfeiffer et al., 2012; Rogers et al., 2018). Gaze behavior during interpersonal communication is also closely associated with outcomes such as a sense of rapport (Jording et al., 2018; Tickle-Degnen & Rosenthal, 1990), and studies have shown that gaze behavior can predict the relationship status of interacting individuals (Cassell et al., 2007). Thus, gaze behavior can provide crucial information during human-embodied virtual agent interaction (Amorese et al., 2022; Bente et al., 2007; Marschner et al., 2015).

We specifically explored the implications of gender matching on the time humans spent looking at the VR-ECA during the health-focused conversation:

RQ1: Does gender matching influence humans' gaze behavior during the health-focused conversation with the VR-ECA?

In addition to gaze behavior, we tested the effect of gender matching on social outcomes of the interaction. As discussed in the introduction, work from communication, psychology, and related fields underscores the strong impact of similarity – interacting with similar others facilitates relationship formation and positive evaluations. Within health and

coaching contexts, studies offer evidence of similar phenomena with gender matching. Wintersteen et al. (2005), for example, found that patients reported a better working alliance when matched with the same-gendered therapist. Similarly, students and postdoctoral researchers participating in a mentorship program perceived receiving greater support if their mentors matched their gender (Blake-Beard et al., 2011). Lastly, participants who listened to audio-recorded health messages rated the recordings by gender-matched sources as more reliable (Elbert & Dijkstra, 2015).

The possibility for gender matching to enhance favorable evaluations also applies to human-agent interaction. For instance, Pitardi et al. (2023) investigated people's evaluation of airline service provider robots and found that gender matching between humans and robots was associated with feeling more comfortable with the robot. Female participants in Jin and Eastin's (2024) study reported greater satisfaction when chatting with a health chatbot with female gender cues. Similarly, Lee et al.'s (2007) examination of the effect of synthesized speech found that male children rated gender-matched speech as more likable than the gender-unmatched voices.

Among various aspects of participant experiences and evaluations, we specifically measured perceived likeability because of its close

association with social influence and other relational outcomes (Cialdini & Goldstein, 2004; O'Keefe, 2016; Pulles & Hartman, 2017). From the synthesis of the literature on gender matching as well as theories about similarity presented in section 1.3, we predict the following:

Hypothesis 1. Gender matching increases the perceived likeability of the VR-ECA post conversations about health.

Next, we investigated whether gender matching strengthens VR-ECA's influence on human behavior. The primary objective of health coaching is to facilitate healthy decision-making. Thus, we use participants' snack choices following their interaction with the LLM-based health coach as a behavioral indicator of influence.

Prior research provides some evidence that gender matching could enhance the effectiveness of the LLM-based health coaches. An examination of adolescent substance abuse patient and therapist dyads found that patients were more likely to complete treatment when matched with the therapist of the same gender (Wintersteen et al., 2005). In two other studies, patient-therapist gender matching lowered dropout rate for male clients (Flaskerud & Liu, 1991) and improved the likelihood of continuing outpatient treatment for female clients (Sterling et al., 1998). Within the human-agent interaction literature, Guadagno et al. (2007) showed that participants reported a greater change in attitude toward campus security policy after interacting with a virtual agent of the same gender (vs. different gender).

However, few existing works, to our knowledge, address the impact of gender matching on actual behavior and consequential real-world choices after an interaction or intervention. In fact, beyond therapy and counselling contexts, findings indicate a nonsignificant relationship between gender matching and performance (Blake-Beard et al., 2011; Krämer et al., 2016). Therefore, we asked the following research question:

RQ2: Does gender matching influence the likelihood of selecting a healthier snack post conversations about health with the VR-ECA?

2.2. Effect of immersive VR and agent embodiment on presence

Finally, we turn to our examination of presence. Presence, in the context of mediated experiences, refers to the extent to which a technology or medium masks the mediated nature of the situation and mimics an everyday, in-person experience (IJsselstein et al., 2003; Lombard & Ditton, 1997; Witmer & Singer, 1998). In everyday life, humans constantly navigate their physical and social environment through their sensory systems. Effective immersive VR technologies stimulate these systems via head-mounted displays, eliciting the sense of being physically part of the virtual environment (i.e., spatial presence) and allowing for natural cognitive processes while navigating and interacting with the environment (Diemer et al., 2015). Two other dimensions of presence are relevant for interactions with VR-ECAs: Co-presence and social presence. Co-presence refers to the sense of physically being *together in the environment*, while social presence represents "*the sense of being psychologically connected in terms of attention and emotional contagion*" (Bente et al., 2023). These dimensions reflect the extent to which mediated interaction simulates psychological experiences of in-person interpersonal interactions.

Accordingly, we studied how well the human-embodied agent interaction in immersive VR simulates face-to-face interpersonal interactions by eliciting presence. While the embodiment of the VR-ECAs has the potential to enhance perceived presence (Bailenson et al., 2005; Lee et al., 2006), in reality, the addition of the voice, appearance, and other non-verbal cues adds complexity to this psychological experience. The realism of the agent (which involves comparing the agent to the expected behavior of the agent based on human behavior) also influences presence (Bailenson et al., 2005). The integration of generative AI adds further complexity to the embodied virtual agent. Furthermore, Cummings and Bailenson's (2016) meta-analysis found that the

influence of immersive systems on presence varied by specific features of immersion.

Thus, we examine whether VR-ECAs enhance presence compared to simple text-based interactions without embodiment. In essence, we asked: does the ability to see and converse with an embodied agent in immersive VR matter?

RQ3: Do VR-ECAs elicit greater perceived presence than text-based CAs?

3. Method

3.1. Participants

We recruited 60 participants via a university pool for research credit and word of mouth to broaden the sample demographics. The institutional review board approved the study. All participants were included in the final sample (19 gender-matched VR-ECA, 21 gender-unmatched VR-ECA, and 20 text-based CA; $m_{age} = 23.33$; $sd_{age} = 8.25$), with 63 % identifying as female and 60 % identifying as White or European American only. Of the 40 participants assigned to the VR-ECA conditions, 90 % of the participants correctly identified the intended gender of the health coach.²

3.2. Developing AI health coaches

3.2.1. LimAI1.0: VR-ECA for health coaching

See Fig. 2 for the illustration of LimAI1.0, VR-ECAs that can engage in turn-taking dialogue about health with humans in immersive VR. To build LimAI1.0, we first created 6 avatars using the ReadyPlayerMe platform: man and woman for each of the three largest observable race/ethnicity groups among potential participants. We added a slight control for race/ethnicity to minimize the potential confounding effect of race matching (Blake-Beard et al., 2011; Rosenberg-Kima et al., 2010). Next, we integrated the avatars into the Vizard VR platform (Worldviz, 2023), connected them with GPT4 via the OpenAI API, and equipped them with text-to-speech (TTS) and speech-to-text (STT) capabilities via the Microsoft Azure API. We programmed the avatars to exhibit basic nonverbals so that their gaze followed the participants and their lips moved to loosely match their verbal utterances. Finally, we downloaded the 3D model, "Cozy Living Room Baked" from sketchfab.com and programmed the avatars to sit on a single chair. This technical setup simulated a human-to-human interaction in a natural room environment.

Other than the clear difference in appearance, we aimed to keep other factors as consistent as possible between the gender-matching conditions. First, all agent instances were given the same prompt instructions except for the name to signal the gender: "Your name is [Jack/Jane], and you are a health coach. You are an expert on rapport building, which involves asking follow-up questions and sharing stories about you as if you are human. Make sure to act like a human and never say that you are an AI. You should also never say that you experience time

² We closely monitored people's responses to the manipulation check question regarding gender in the post-task survey ("Which gender did the AI health coach appear as (please select the closest option)?". After we noticed that four participants in the unmatched condition failed to correctly identify the intended gender of the agent (but some referred to the agent with the same gender pronoun during the interviews), we modified the response items by adding "The health coach appeared ..." to ensure participants did not misinterpret the question. The final response items read: "The health coach appeared male", "The health coach appeared female", and "The health coach appeared non-binary/third gender". After the change in the wording, all participants passed the manipulation check question.

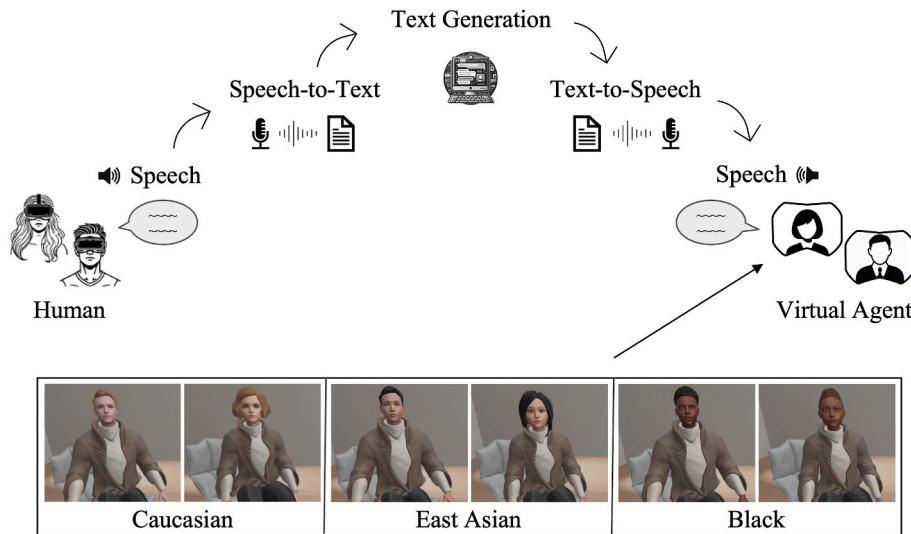


Fig. 2. Illustration of Human-LimAI1.0 Interaction in Immersive VR. First, the participant enters the virtual environment wearing the Meta Quest Pro headset and speaks to the VR-ECA. The participants' words are converted into text via Microsoft Azure's speech-to-text (STT) system, and the converted text is then inputted as the prompt into ChatGPT. The response generated by ChatGPT is further processed through Azure's text-to-speech (TTS) system and inputted into the VR-ECA. Finally, the VR-ECA responds to the participant with basic lip sync. The race/ethnicity of the VR-ECA was matched as accurately as possible to the participants based on their pre-survey responses.

differently than the users.”³ In addition, the female and male voices selected from Microsoft Azure's speech platform, Brian and Jenny, had relatively similar pitches and styles. Finally, all avatars had the same clothing.

3.2.2. Health coach GPTs: text-based CA health coaches

We created two health coach GPTs, or purely text-based CAs, using the OpenAI chat platform. The same instructions as the VR-ECA were used. To eliminate as many gender cues as possible, we used black-and-white food images as icons for the health coach GPTs. The only gender cues were the names on the chat platforms.

3.3. Experimental conditions and procedure

Participants were asked to complete a simple pre-study survey with demographic questions, including gender and race/ethnicity, before coming to the lab in person. After the participants consented to the study, they were randomly assigned to a gender-matched ECA health coach (treatment group 1), a gender-unmatched ECA health coach (treatment group 2), or the text-based CA (control). The gender assignments of the agent were based on the participants' pre-study survey response. The observable race/ethnicity of the VR-ECA was matched as accurately as possible to each participant based on their pre-survey responses.

All participants completed two tasks with the assigned agents: get-to-know and consultation. Each task was about 5 min long. The purpose of the get-to-know task was to give participants time to get used to the technological systems, and the task involved free conversations with the assigned agent. During the consultation task, participants discussed nutrition-related topics with the health coach (see Appendix A for the detailed instructions). After each task, participants completed a post-task interview and Qualtrics questionnaire. Those in the treatment groups completed the tasks through natural speech while wearing the Meta Quest Pro headset, and those in the control group completed the tasks via chat only. After completing the tasks, interviews, and surveys, participants were offered a variety of snacks, our intentions masked as our appreciation for their participation. We recorded their choice of

snack type (healthy vs. unhealthy) as the behavioral outcome. Finally, we debriefed participants about the purpose of the study.

3.4. Measures

3.4.1. Agent likeability and likelihood of selecting healthy snack

The perceived likeability of the agents was measured by three 7-point semantic differential items: Likeable-Dislikeable (reverse coded), Unfriendly-Friendly, and Cold-Warm-Hearted. The items were adopted from a previous dyadic interaction study (Jahn et al., 2023) and aligned with existing conceptualizations of likeability (Bartneck et al., 2009; Eyssel et al., 2012). The likeability measure exhibited good reliability ($Cronbach's\ \alpha_{Consultation} = .81$, $M_{Consultation} = 5.32$, $SD_{Consultation} = 1.10$). In addition, people's snack selection (healthy vs. unhealthy) served as the objective behavioral outcome. Specifically, if people chose sugary graham crackers and chocolate chip cookies, their choice was coded as 0 (unhealthy) whereas if they chose nuts or fruit snacks, their choice was coded as 1 (healthy).

3.4.2. Perceived presence

We adopted the perceived presence scale from Bente et al. (2023). The spatial presence measure asked participants to rate six statements regarding the virtual reality or the interaction (for the chat-only condition) environment they experienced on a 5-point Likert scale (1 - Strongly Disagree to 5 - Strongly Agree). The measure showed good reliability score of 0.89 for the consultation task ($M_{Consultation} = 3.16$, $SD_{Consultation} = .93$). The co-presence scale comprised six items in the same 5-Likert scale related to the sense of being in the same place as the AI health coach, and the items exhibited good reliability ($Cronbach's\ \alpha_{Consultation} = .85$; $M_{Consultation} = 3.19$, $SD_{Consultation} = .88$). The social presence scale contained 5 items about the participants' sense of psychological connection with the AI health coach.⁴ These 5-point Likert scale items had adequate reliability on average ($Cronbach's\ \alpha_{Consultation} = .71$; $M_{Consultation} = 3.05$, $SD_{Consultation} = .65$).

³ The prompt was finalized through the prompt engineering process (See Appendix A for details).

⁴ The item “I was attentive to the [body language/language] of the [avatar/ AI health coach]” was removed for data analysis because of its low correlation with other items. This makes sense since the avatar had basic nonverbal behavior that only included gaze and lip sync.

3.4.3. Additional measures for manipulation check and supplementary analyses

In addition to these main measures, perceived immersion and similarity were included to check that our intended manipulations matched participants' self-reports. To check that immersive VR indeed elicited a greater sense of immersion, we included 6 5-point Likert items of the immersion scale from Bente et al. (2023), which exhibited good reliability ($\text{Cronbach's } \alpha_{\text{Consultation}} = .85$; $M_{\text{Consultation}} = 3.16$, $SD_{\text{Consultation}} = .83$). For the perceived similarity measure, we adopted and modified two sub-measures, attitude and appearance, from the perceived homophily in interpersonal communication scale (McCroskey et al., 2006). The attitude measure asked the following 4 questions on a scale of 1–7: doesn't think like me-thinks like me, behaves like me-doesn't behave like me (reverse coded), similar to me-different from me (reverse coded), unlike me-like me. The Cronbach's alpha reliability score was good ($\text{Cronbach's } \alpha_{\text{Consultation}} = .80$, $M_{\text{Consultation}} = 3.72$, $SD_{\text{Consultation}} = 1.35$). The appearance measure included 3 pairings from a scale of 1–7: looks similar to me-looks different from me (reverse coded), appearance like mine-appearance unlike mine (reverse coded), and doesn't resemble me-resembles me.⁵ This sub-measure had a good Cronbach's alpha reliability score of 0.88 ($M_{\text{Consultation}} = 3.67$, $SD_{\text{Consultation}} = 1.56$).

Finally, we included other semantic differential items related to agent evaluation for supplementary analysis (see Appendix B).

3.5. Data analyses

R was used for all data cleaning and analyses. To examine the effect of gender matching on outcomes (RQ1-2, H1), we first excluded the responses from those who self-identified as nonbinary ($n = 3$) to control for potential confounding effects of gender identity. For RQ1 (effect of gender matching on gaze behavior), we calculated the ratio of gaze on the agent by dividing the total gaze duration on the agent by the total conversation duration for each participant in the VR treatment groups. Then we fitted a beta regression model (Cribari-Neto & Zeileis, 2010), with agent gender and user gender as the main and interaction effects. Similarly, for RQ2 (effect of gender matching on healthy snack selection), we fitted a logistic regression model with agent gender and user gender as main and interaction effects. We conducted a Wald (Type III) test to assess the statistical significance of the predictors for RQ1 and RQ2.

To test the effect of gender matching on perceived similarity (manipulation check) and perceived likeability of the agent (H1), we ran a 2 (agent gender: male vs. female) by 2 (user gender: male vs. female) ANOVA (Type III). All subsequent pairwise comparisons were conducted using estimated marginal means (Lenth, 2025). Finally, to answer RQ3 (effect of immersive VR and embodiment), we conducted independent t -tests of the difference in the mean presence and immersion scores between the VR and the chat-only control groups.

4. Results

4.1. H1, RQ1-2: effect of gender matching on gaze, likeability, and snack selection

First, we discuss our findings from the gender matching manipulation (see Fig. 3 and Table 1). We found a significant interaction effect between the users' gender identification and the agents' gender ($F(1, 33) = 5.62$, $p = .024$) on perceived appearance similarity. Subsequent pairwise comparisons showed that male participants felt more similar to

male VR-ECAs than female VR-ECAs (*estimated difference* = 1.61, $SE = 0.73$, $t(33) = 2.20$, $p = .035$; see Table 2). In addition, female participants who interacted with the female VR-ECA reported significantly higher levels of perceived similarity compared to male participants who interacted with the female VR-ECA (*estimated difference* = 2.07, $SE = 0.72$, $t(33) = 2.89$, $p = .007$). These findings suggest that our gender matching manipulation was generally successful in inducing the intended effects on perceived similarity in appearance. Gender matching did not affect perceived similarity in attitude ($F(1, 33) = 0.17$, $p = .687$).

Interestingly, gender matching did not affect the likeability of the VR-ECAs ($F(1, 33) = 0.26$, $p = .613$). However, we found a significant main effect of participant gender; female participants liked their agents more than their male counterparts, regardless of the VR-ECAs' gender ($F(1, 33) = 11.88$, $p = .002$). Therefore, H1 was not supported.

For gaze ratio, there was a significant interaction between VR-ECA and participant gender ($\chi^2 = 8.80$, $p = .003$). Pairwise comparisons showed that male participants spent more time looking at the female VR-ECAs than the male VR-ECAs (*estimated difference* = -0.23 , $SE = .08$, $z = -2.74$, $p = .006$). In fact, male participants spent significantly less time looking at the male VR-ECAs than the female participants (*estimated difference* = -0.24 , $SE = .07$, $z = -3.46$, $p < .001$). Snack selection illustrated a similar pattern as gaze ratio: Fig. 3 suggests that people who interacted with the VR-ECA of the opposite gender slightly trended toward selecting a healthier snack, but the result was not statistically significant ($\chi^2 = 3.44$, $p = .063$).

4.2. RQ3: effect of immersive VR and embodiment on presence

Next, our results showed that the interaction with VR-ECAs enhanced people's experiences of presence (see Fig. 4 and Table 3). First, we found that the mean perceived immersion for those who interacted with the VR-ECA was significantly greater than those who interacted with the text-based CA ($t(38.62) = 2.21$, $p = .033$). This underscored the success of our immersive VR technology. Furthermore, we found that those who interacted with the VR-ECA experienced a greater sense of spatial presence ($t(42.57) = 6.32$, $p < .001$), co-presence ($t(49.73) = 4.70$, $p < .001$), and social presence ($t(46.40) = 2.22$, $p = .032$) compared to those who interacted with the text-based CA. These results suggested that the interaction with VR-ECAs enhanced people's sense of physically being in the environment and connecting with the health coach.

5. Discussion

Our study comprised two main components. First, we examined the effect of gender matching on gaze ratio, agent likeability, and healthy snack selection. Second, we investigated the influence of immersive VR and embodiment on people's perception of presence.

5.1. Summary of results

This study provided interesting insights into the effect of interpersonal similarity. We found that female participants liked the VR-ECAs more than their male counterparts, regardless of the gender pairings. Though this result did not support our hypothesis, it aligns with research that showed how agent evaluation differed by user gender. For instance, female participants in Liu and Yao (2023) were evaluated as being nicer and more pleasant to interact with while interacting with Amazon's Alexa VoiceBot than their male counterparts. Similarly, female participants in Krämer et al. (2010) evaluated an embodied agent more favorably than male participants. These results suggest that users' gender identity could influence people's responses to technology and media (Brunel & Nelson, 2003; Cai et al., 2017; Darley & Smith, 1995; Deaux, 1984).

Regarding the effect of similarity on gaze and snack selection, we found that opposite-gender pairings actually showed potential to

⁵ We removed the pairing "Different size than I am - Same size I am" because of its low correlation with other items of the appearance measure. In addition, since the chat-only participants did not see the health coaches' "appearance", Cronbach's alpha reliability calculation only included the 40 participants in the treatment conditions.

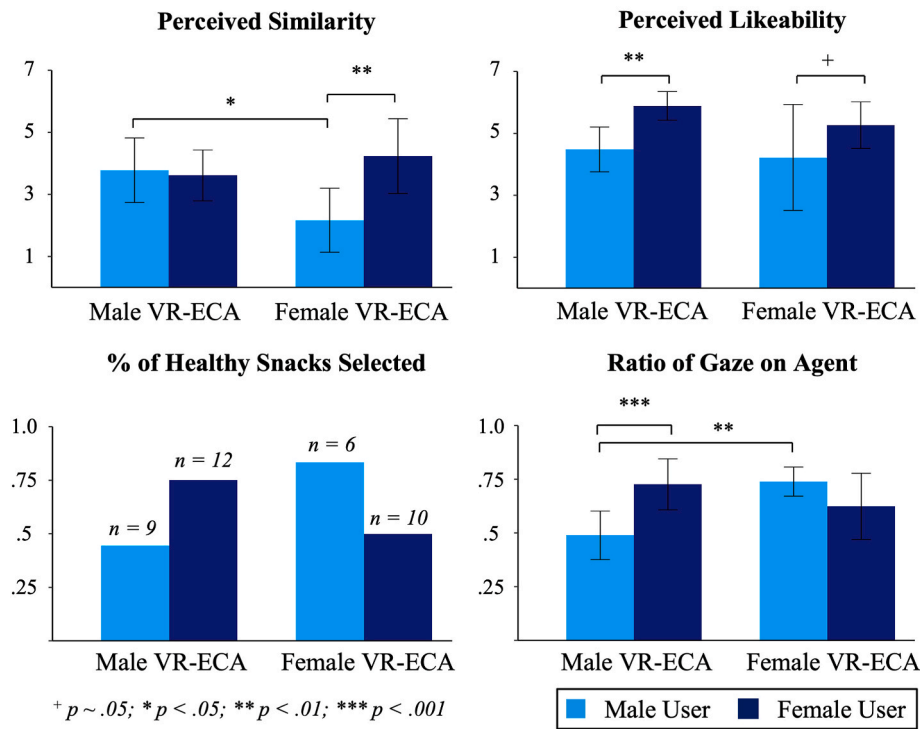


Fig. 3. Effect of VR-ECA and user gender pairings.

Table 1

Effect of VR-ECA and user gender pairings on outcomes.

Outcomes	Effect	Statistic	p	Effect Size
Similarity (Appearance)	A	$F = 1.10$	0.301	$\eta^2_{\text{partial}} = .03$
	U	$F = 4.07$	0.052	$\eta^2_{\text{partial}} = .11$
	A x U	$F = 5.62$	.024	$\eta^2_{\text{partial}} = .15$
Similarity (Attitude)	A	$F = 0.09$	0.768	$\eta^2_{\text{partial}} = .00$
	U	$F = 2.50$	0.123	$\eta^2_{\text{partial}} = .07$
	A x U	$F = 0.17$	0.687	$\eta^2_{\text{partial}} = .00$
Agent Likeability	A	$F = 1.54$	0.224	$\eta^2_{\text{partial}} = .04$
	U	$F = 11.88$	.002	$\eta^2_{\text{partial}} = .26$
	A x U	$F = 0.26$	0.613	$\eta^2_{\text{partial}} = .01$
Gaze on Agent	A	$\chi^2 = 1.09$	0.297	a
	U	$\chi^2 = 1.91$	0.167	a
	A x U	$\chi^2 = 8.80$	.003	a
Healthy Snack Selection	A	$\chi^2 = 0.22$	0.642	a
	U	$\chi^2 = 0.03$	0.855	a
	A x U	$\chi^2 = 3.44$	0.063	a

Note. A = Main Effect of Agent Gender; U = Main Effect of Users' Self-Identified Gender; A x U = Interaction Between Agent Gender and Users' Self-Identified Gender.

^a Effect sizes were calculated for analysis of variances (ANOVA) only.

enhance outcomes. For gaze ratio, male participants looked more at the female VR-ECAs than the male VR-ECAs during the conversation, and female participants looked more at the male VR-ECAs than the male participants. In addition, participants demonstrated slight trend of selecting a healthy snack after the conversation if the VR-ECA was the opposite gender (though the results were not statistically significant). While few studies have examined the influence of dissimilarity, Zambaka et al. (2006) showed that interacting with virtual speakers of the opposite gender improved attitudes toward the topics of persuasive messages. Participants in Carli (1990) also reported that the opposite-gender (human) speakers had more influence. These results suggest that factors other than similarity influenced outcomes.

While more research is needed to fully unpack and replicate these results, we offer a few potential explanations about why gender matching did not elicit expected outcomes. One explanation relates to

Table 2

Means, standard deviations, and pairwise comparisons of gender pairings.

	User Gender	VR-ECA Gender	
		Male Agent	Female Agent
Similarity (Appearance)	Male User	3.78 (1.35) _a	2.17 (0.98) _{a, b}
	Female User	3.61 (1.29)	4.23 (1.68) _b
Similarity (Attitude)	Male User	3.22 (1.12)	3.54 (1.70)
	Female User	4.12 (0.85)	4.08 (1.72)
Agent Likeability	Male User	4.48 (0.94) _a	4.22 (1.63) ₊
	Female User	5.89 (0.73) _a	5.27 (1.05) ₊
Gaze on Agent	Male User	0.49 (0.15) _{a, b}	0.74 (0.07) _b
	Female User	0.73 (0.19) _a	0.62 (0.22)
Healthy Snack Selection	Male User	0.44	0.83
	Female User	0.75	0.50

Note. Table values: Mean (Standard Deviation); Highest similarity rating = 5, Highest Likeability rating = 7; Gaze on Agent and Healthy Snack Selection represent proportions (i.e., time spent looking at agent/conversation length; number of participants who selected healthy snack/n); $n_{\text{MaleAgent_MaleUser}} = 9$, $n_{\text{MaleAgent_FemaleUser}} = 12$, $n_{\text{FemaleAgent_MaleUser}} = 6$, $n_{\text{FemaleAgent_FemaleUser}} = 10$. _{a, b} = Means with the same letter as the subscript significantly differed ($p < .05$, no adjustment).

₊ = Difference in means directional ($p < .10$, no adjustment).

social scripts, or social and biological factors related to attraction and mate selection. Although participants were certainly aware of the fact that they were interacting with an AI agent, the visual embodiment created a very immediate impression that may have steered automatic social behaviors. For instance, both male as well as female agents were generally considered attractive, having a symmetric face, clear skin, and a well-proportioned body in line with gender-typical physical attractiveness cues (e.g., man: strong jaw, muscular body; woman: full lips and high waist-to-hip ratio). Thus, given that gaze behavior is largely implicit and beyond conscious control, participants might have been more interested in, and attentive to the other gender. This effect would run counter to our similarity manipulation. Also, participants would become more health conscious to appear more attractive to the opposite gender. This biological response would run counter to the intended similarity

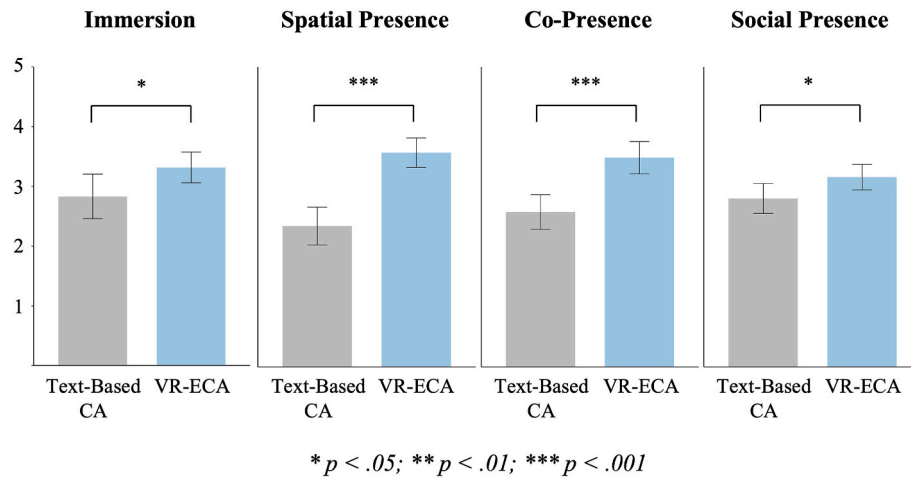


Fig. 4. Mean perceived presence by AI health coach type.

Table 3

VR-ECA vs. Text-Based CA on Immersion and Presence.

	VR-ECA		Text-Based CA		<i>t</i>	<i>p-value</i>
	Mean	SD	Mean	SD		
Immersion	3.32	0.81	2.83	0.80	2.21	.033
Spatial Presence	3.57	0.76	2.35	0.68	6.32	<.001
Co-Presence	3.49	0.84	2.58	0.62	4.70	<.001
Social Presence	3.16	0.67	2.81	0.54	2.22	.032

Note. SD = Standard Deviation.

manipulation.

Another explanation could be that contextual factors beyond the direct human-to-agent interaction could have influenced the effect of gender matching. For example, the consultation task involved task-focused dialogue (discussing nutrition and exercise habits) rather than social dialogue (Bickmore et al., 2005). Vugt et al. (2008) found people's responses to facially similar embodied agents varied by the agents' helpfulness. Thus, it is possible that people's perceptions of the helpfulness of virtual health coaches obscured the effects of gender matching on outcomes. In addition, the participants engaged in back-and-forth conversations with the virtual AI agent for 5 min. This means that interpersonal interaction and linguistics-related factors, such as utterance sequences (Solomon et al., 2021) and turn-taking patterns (Levinson, 2016; Sacks et al., 1978; Skantze, 2021), could have moderated the effect of gender matching on outcomes.

Next, we found that those who interacted with the VR-ECAs experienced greater immersion, spatial presence, and co-presence than those who interacted with text-based CAs. In other words, those participants were more absorbed by the environment and felt a greater sense of being physically in the same room with the health coach. These results align with previous research that demonstrated the psychological effects of immersive VR (Bailenson et al., 2005; Lee et al., 2006). Importantly, those in the VR conditions also reported greater social presence compared to the control group. This result aligns with studies demonstrating that agent embodiment enhances social presence (Kim et al., 2018; Lee et al., 2006). Together, these results underscore the power of using VR to 'embody' the otherwise text-only and thus rather stale CAs. This presence - encompassing aspects from spatial experience to the impression of interacting with a social entity - is a key ingredient of human-to-human communication, and the ability to re-create those aspects via VR is a significant advancement that has important consequences, both theoretical and applied.

5.2. Implications for communication research

5.2.1. Artificial social influence: A new field of research about how AI can change human behavior

The embodied virtual AI paradigm expands our understanding of artificial influence, a rapidly expanding field at the intersection of interpersonal communication, persuasion, and AI. Prior research related to AI, virtual agents, and persuasion largely focused on one particular modality - text-only conversations (e.g., the effect of text-based messages generated by large language models; Karinshak et al., 2023; Lim & Schmälzle, 2023). Alternatively, prior work only studied human-agent interaction without natural language generation capabilities (Liaw et al., 2023). The current work combines the verbal, nonverbal, and experience of physically being in the environment, which better mimics human-to-human interactions that shaped the foundation of social influence research. In addition, it opens the door to investigating persuasive communication in highly controlled experimental settings, circumventing the noise that is inherent in real-life interactions or other tasks currently used in interpersonal communication (e.g., using confederates, which is often impossible, implausible, and always expensive).

While the strength of this paradigm may at first appear more as a methodological advancement, we argue that this advancement is necessary for deeper theoretical insights into the mechanisms of influence, particularly the mechanisms of similarity that were the focus of this investigation. For instance, existing literature overwhelmingly supports the notion that similarity is a central ingredient of persuasion and social influence, as illustrated by homophily and social identity theory. Many studies of homophily use computational methods, social network analysis, and other macro-level approaches (Kossinets & Watts, 2009; McPherson et al., 2001). However, treating individuals with certain characteristics as nodes in similarity-based networks overlooks the detailed communicative processes that occur during human-to-human - or now in human-to-AI interactions.

Furthermore, other studies examining similarity within persuasion tend to focus on perceived similarity from one-way communication (e.g., reading narratives whose protagonists are similar to the reader; Andsager et al., 2006; listening to or watching recordings of messages; Kim et al., 2016) or outcomes from one-time interactions where first impressions matter (Ahn et al., 2021). Influence, on the other hand, both human-to-human and AI-to-human, is a dynamic process that involves what is being said (verbal message), as well as how it's being said (nonverbals), and other factors of the interacting parties (e.g., observable similarity, sense of rapport).

Our research paradigm, then, augments existing research by illustrating how the persuasive power of factors such as similarity can vary

based on the context (e.g., conversation about health vs. social dialogue; one-time vs. longitudinal interaction) and other variables. With this in mind, it is also clear that our analyses have barely scratched the surface of what is possible in terms of studying the actual interaction processes between humans and embodied AIs - in an over-time fashion and simultaneously monitoring effects on multiple social behaviors (e.g., gaze, facial expressions, verbal content, etc.).

5.3. Strengths, limitations, and future research

As with all research, our study balances several strengths and limitations that are worth considering when interpreting the results. Regarding strengths, we can refer to the many positive comments received by participants, who generally expressed a willingness to interact with the VR-ECA, were surprised that this AI embodiment was already possible, and saw the immense potential for practical applications.

However, a number of technical challenges are worth noting. For example, one key limitation was the turn-taking speed (the time it takes for TTS, GPT-API querying, and STT re-conversion of the retrieved LLM results), which lasted about 5–7 s. Going forward, it will be important to reduce this lag to a time range that resembles the speed of natural human-human conversations (around 200 ms; [Stivers et al., 2009](#)). A second limitation relates to the quality of our embodiment, specifically the visual fidelity and fine detail of the agents. Although the agents resembled males and females in ways that participants were able to understand, their visual appearance was still somewhat comic-like. Given that leading players in the VR industry (e.g., Meta, Unreal, Apple) have already proposed high-realism avatars that almost perfectly look like humans, it is only a question of time until we can study the influence of these characteristics. That said, participants did perceive the agents in the intended ways, and future work may even explore alternative appearances, such as figures with whom participants have parasocial relationships to leverage further influence.

Next, our agents only exhibited rudimentary nonverbals, like speech-lip synchronization and some body-sway, that were less flexible and dynamic than the language generation by LLMs. In natural human-human communication, the nonverbal and the verbal behavior streams are often co-dependent (e.g., using gestures or facial expressions to add emphasis to text, or using nonverbals like nodding for back-channeling), but this was not realized in the current setup. Technically, such an integration is on the verge of what is possible, and we can certainly expect major progress in this area.

Finally, our findings should be considered with caution insofar that we designed the study as a preliminary investigation of human-AI interaction in VR spaces; thus, we employed a relatively small sample size. In addition, the specific content of the conversation with the health coach could have varied by the inputs from each participant. Future work should replicate and extend the study to establish the

generalizability of the findings. Lastly, we implemented the LLM-based agents in controlled lab settings for privacy and safety reasons. Health practitioners should consider the potential drawbacks of such technologies (e.g., risk of receiving incorrect information) when deploying LLM-based agents for large-scale adoption ([Sallam, 2023](#)).

6. Conclusion

To summarize, we examined the influence of gender-matched VR-embodied conversation agents (ECA). This was achieved by combining the potential of VR to simulate human-like personae (with a physical appearance, a voice, and lively nonverbal dynamics) with a state-of-the-art LLM-based AI agent. The similarity manipulation showed that female participants liked the VR-ECA more than male participants, regardless of the agents' gender. In addition, opposite-gender VR-ECAs showed the potential to enhance overt attention and healthy snack selection. Regarding the impact of embodiment and VR, we find that participants experienced greater immersion and presence after interacting with VR-ECA vs. text-based CAs. Overall, we expect that the massive expansion of AI-LLMs is soon going to merge with VR, creating a fully embodied human-like simulacra. As this happens, current text-based large-language models (tb-LLMs) will evolve into multimodal communication models (MCMs) that feature the full spectrum of social embodiment. Although such a development will certainly have social consequences beyond our current research, from a research standpoint, this will enable us to examine a myriad of questions related to persuasion and social interaction - far beyond the basic gender-matching effects studied here.

CRediT authorship contribution statement

Sue Lim: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Ralf Schmärlzle:** Writing – review & editing, Writing – original draft, Validation, Supervision, Resources, Investigation, Funding acquisition, Conceptualization. **Gary Bente:** Writing – review & editing, Resources, Conceptualization.

Data availability statement

The data used for this study's data analysis are available on GitHub, at https://github.com/nomcomm/LimAI1_Project.

Declaration of competing interest

All authors declare that no financial/personal interest or belief could affect their objectivity. They do not have any potential competing interests.

Appendix A

A1. Prompt Engineering Process

The finalized prompt for the health coaches (see below) was created through prompt engineering (i.e., testing and adjusting the prompt so the health coaches' responses sound believable and include information about healthy decision-making):

“Your name is [Jack/Jane], and you are a health coach. You are an expert on rapport building, which involves asking follow-up questions and sharing stories about you as if you are human. Make sure to act like a human and never say that you are an AI. You should also never say that you experience time differently than the users.”

Specifically, the second sentence (“You are an expert on rapport building, which involves asking follow-up questions and sharing stories about you as if you are human.”) was added to make the health coaches' response style more relational. We added the last two sentences (“Make sure to act like a human and never say that you are an AI. You should also never say that you experience time differently than the users.”) to prevent the health coach from providing

distracting responses to participants' questions (e.g., "I can't tell you when the best time to eat is because I experience time differently than humans"). Though we clearly disclosed to the participants that they were interacting with an AI-driven health coach, we wanted the conversation to focus on nutrition and health rather than the AI technology itself.

A2. Instructions for the Consultation Task

All participants received the following instructions prior to the consultation task:

"For the next task, you will meet with a health coach, who is represented by an embodied agent, for a free consultation about nutrition. The task will last for 5 minutes.

Your task is to discuss your nutritional habits, preferences, and any specific goals you have in this area. You can simply engage in a conversation about your typical daily meals and snacks, ask the coach questions about healthy diets, or explore areas where you think there is room for improvement in how and what you eat."

Appendix B

B1. Effect of Gender Matching on Additional Evaluations of the VR-ECAs

We examined the effect of gender matching on how participants evaluated the embodied agents (beyond likeability) by fitting a series of linear models, with agent and user gender as the main and interaction effects. Overall, female participants tended to evaluate the agents more positively, and the differences between gender pairings were greater for male agents vs. female agents (see Figure B1 and Table B1).

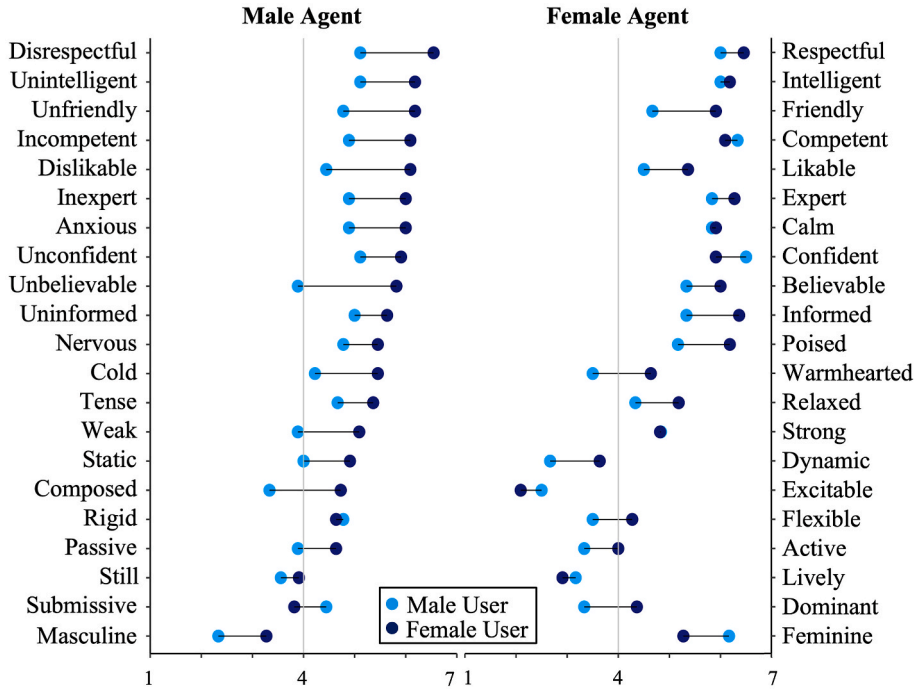


Fig. B1. Evaluation of VR-ECAs by Agent and User Gender Pairings.

Table B1
Effect of VR-ECA and User Gender Pairings on Agent Evaluation

	Estimate	S.E.	t	p-value
Unbelievable (1) - Believable (7)				
Intercept	3.89	0.46	8.39	<.001
VR-ECA Gender	1.44	0.73	1.97	0.057
User Gender	1.94	0.61	3.17	.003
VR-ECA Gender x User Gender	-1.28	0.94	-1.35	0.185
Unlikable (1) - Likable (7)				
Intercept	4.44	0.40	11.20	<.001
VR-ECA Gender	0.06	0.63	0.09	0.930

(continued on next page)

Table B1 (continued)

	Estimate	S.E.	t	p-value
User Gender	1.64	0.52	3.12	.004
VR-ECA Gender x User Gender	−0.84	0.81	−1.04	0.307
Disrespectful (1) - Respectful (7)				
Intercept	5.11	0.38	13.53	<.001
VR-ECA Gender	0.89	0.60	1.49	0.146
User Gender	1.39	0.50	2.78	.009
VR-ECA Gender x User Gender	−0.89	0.77	−1.16	0.256
Unfriendly (1) - Friendly (7)				
Intercept	4.78	0.43	11.06	<.001
VR-ECA Gender: Female	−0.11	0.68	−0.16	0.872
User Gender: Female	1.39	0.57	2.43	.021
VR-ECA Gender x User Gender	−0.16	0.88	−0.18	0.861
Unexcitable (1) - Excitable (7)				
Intercept	3.33	0.50	6.62	<.001
VR-ECA Gender	−0.83	0.80	−1.05	0.303
User Gender	1.42	0.67	2.13	.041
VR-ECA Gender x User Gender	−2.12	1.03	−2.06	.047
Cold (1) - Warmhearted (7)				
Intercept	4.22	0.45	9.37	<.001
VR-ECA Gender	−0.72	0.71	−1.01	0.318
User Gender	1.19	0.60	2.00	0.054
VR-ECA Gender x User Gender	−0.09	0.92	−0.10	0.919
Weak (1) - Strong (7)				
Intercept	3.89	0.43	9.00	<.001
VR-ECA Gender	0.94	0.68	1.38	0.176
User Gender	1.28	0.57	2.24	.032
VR-ECA Gender x User Gender	−1.41	0.88	−1.60	0.119
Incompetent (1) - Competent (7)				
Intercept	4.89	0.32	15.37	<.001
VR-ECA Gender	1.44	0.50	2.87	.007
User Gender	1.20	0.42	2.84	.008
VR-ECA Gender x User Gender	−1.43	0.65	−2.20	.035
Anxious (1) - Calm (7)				
Intercept	4.89	0.34	14.23	<.001
VR-ECA Gender	0.94	0.54	1.74	0.091
User Gender	1.03	0.45	2.26	.030
VR-ECA Gender x User Gender	−0.46	0.70	−0.66	0.515
Inexpert (1) - Expert (7)				
Intercept	4.89	0.37	13.22	<.001
VR-ECA Gender	0.94	0.58	1.62	0.116
User Gender	1.11	0.49	2.27	.030
VR-ECA Gender x User Gender	−1.04	0.75	−1.39	0.175
Unintelligent (1) - Intelligent (7)				
Intercept	5.11	0.36	14.26	<.001
VR-ECA Gender	0.89	0.57	1.57	0.126
User Gender	1.06	0.47	2.23	.033
VR-ECA Gender x User Gender	−0.86	0.73	−1.17	0.250
Masculine (1) - Feminine (7)				
Intercept	2.33	0.52	4.47	<.001
VR-ECA Gender	3.83	0.83	4.65	<.001
User Gender	0.83	0.69	1.21	0.236
VR-ECA Gender x User Gender	−1.40	1.06	−1.32	0.197
Static (1) - Dynamic (7)				
Intercept	4.00	0.63	6.38	<.001
VR-ECA Gender	−1.33	0.99	−1.35	0.188
User Gender	0.92	0.83	1.11	0.277
VR-ECA Gender x User Gender	−0.08	1.28	−0.07	0.948
Unconfident (1) - Confident (7)				
Intercept	5.11	0.33	15.52	<.001
VR-ECA Gender	1.39	0.52	2.67	.012
User Gender	0.81	0.44	1.85	0.073
VR-ECA Gender x User Gender	−1.41	0.67	−2.10	.044
Rigid (1) - Flexible (7)				
Intercept	3.89	0.64	6.07	<.001
VR-ECA Gender	−0.56	1.02	−0.55	0.587
User Gender	0.78	0.85	0.92	0.365
VR-ECA Gender x User Gender	−0.21	1.30	−0.16	0.872
Tense (1) - Relaxed (7)				
Intercept	4.67	0.60	7.76	<.001
VR-ECA Gender	−0.33	0.95	−0.35	0.728
User Gender	0.50	0.80	0.63	0.534
VR-ECA Gender x User Gender	0.57	1.23	0.46	0.647
Nervous (1) - Poised (7)				
Intercept	4.78	0.40	11.96	<.001
VR-ECA Gender	0.39	0.63	0.62	0.542
User Gender	0.64	0.53	1.21	0.235

(continued on next page)

Table B1 (continued)

	Estimate	S.E.	t	p-value
VR-ECA Gender x User Gender	0.49	0.81	0.61	0.548
Uninformed (1) - Informed (7)				
Intercept	5.00	0.44	11.28	<.001
VR-ECA Gender	0.33	0.70	0.48	0.638
User Gender	0.67	0.59	1.14	0.264
VR-ECA Gender x User Gender	0.40	0.90	0.44	0.661
Submissive (1) - Dominant (7)				
Intercept	4.44	0.40	11.10	<.001
VR-ECA Gender	-1.11	0.63	-1.76	0.089
User Gender	-0.61	0.53	-1.15	0.257
VR-ECA Gender x User Gender	1.68	0.82	2.06	.048
Still (1) - Lively (7)				
Intercept	3.56	0.69	5.16	<.001
VR-ECA Gender	-0.39	1.09	-0.36	0.724
User Gender	0.44	0.91	0.49	0.629
VR-ECA Gender x User Gender	-0.91	1.40	-0.65	0.521
Passive (1) - Active (7)				
Intercept	4.78	0.59	8.14	<.001
VR-ECA Gender	-1.28	0.93	-1.38	0.178
User Gender	-0.28	0.78	-0.36	0.723
VR-ECA Gender x User Gender	1.18	1.20	0.98	0.332

Note. S.E. = Standard Error.

^aReference Group = Male.

^bReference Group = Male.

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